$\mathbb{Z}^\times = \{ n \in \mathbb{Z} : \exists n^{-1} \in \mathbb{Z} \text{ s.t. } n \cdot n^{-1} = 1 \}$

$= \{ 1, -1 \} = \{ \pm 1 \}$.

We can prove directly that $(\{\pm 1\}, \cdot)$ is Abelian group.

$\cdot$	1	-1
1	1	-1
-1	-1	1

Kally Table

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Recall: Let G be a set $*: \overbrace{G \times G \to G}^{\text{closure}}$ a binary opperation. We say G is a group if

1. $(a*b)*c = a*(b*c)$, $\forall a,b,c \in G$. } Associative
2. $\exists e \in G$ s.t. $a*e = e*a = a$, $\forall a \in G$ } Identity

Inverses { 3. $\forall a \in G$, $\exists a^{-1} \in G$ s.t. $a * a^{-1} = a^{-1} * a = e$.
($\neq \frac{1}{a}$ or not necessarily)

commutativity { If $a*b = b*a$, $\forall a,b \in G$, we say G is Abelian.

Thm. 2.1 In a group, there is a unique identity element.

Proof Suppose e and e' both satisfy the identity condition.

Let $a \in G$. We have $a*e = a*e' = a$.

(Specifically, $a*e = e*a = a$ and $a*e' = e'*a = a$ both hold. More specifically, $a*e = a$ and $a*e' = a$ gives us $\underline{a*e = a*e'}$ by transitivity.) The Ace in our sleeves is multiplying by inverses. We group operation on the left by a^{-1}: $a^{-1}*(a*e) = a^{-1}*(a*e')$.

By Associativity, $(a^{-1}*a)*e = (a^{-1}*a)*e'$.

By inverses, $e*e = e*e'$. Since e is an identity, $e = e'$. Thus, <u>the</u> identity element is unique. ∎

We will drop the asterisk and write abstract groups multiplicatively. Thus $a*b$ becomes ab and $b*b^{-1}$ becomes bb^{-1}. e, not 1, will still be the identity.

Thm 2.2: In a group, right & left cancellation hold.

i.e. $ab = ac \Longrightarrow b = c$ and $ba = ca$

$\Longrightarrow b = c$

Proof: Suppose $ab = ac$. We group opperation on the left by a^{-1}. Thus $a^{-1}(ab) = a^{-1}(ac)$.

Hence, $\underbrace{(a^{-1}a)b = (a^{-1}a)c}_{\text{by Assoc.}}$, so $\underbrace{eb = ec}_{\text{by inverses}}$,

and thus $\underbrace{b = c}_{\text{by identity condition}}$.

(proving left hand cancelation is similar)

Thm 2.3. Inverses are Unique

Proof: Let $a \in G$ and suppose a^{-1} & b both act as inverses. Then, $aa^{-1} = ab = e$.

Then we group opp. on left by an inverse of a, call it c, $c(aa^{-1}) = c(ab)$. Thus, Assoc.

$(ca)a^{-1} = (ca)b$. Hence, $ea^{-1} = eb$, and

identity $a^{-1} = b$. by Thm 2.2 inverses

$\blacksquare$